Physical Chemistry Cume March 2 2018

1) Derive an expression for the <u>steady state rate</u> of the sequence of reactions

$$A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} B \underset{k_{-2}}{\overset{k_2}{\rightleftharpoons}} C \underset{k_{-3}}{\overset{k_3}{\rightleftharpoons}} D$$

With [A] maintained at a fixed value and the product D removed as soon as it is formed.

2) Some enzyme reactions involving two substrates A and B occur by the following mechanism (called the ordered ternary complex mechanism):

$$E + A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} EA$$

$$EA + B \overset{k_2}{\rightarrow} EAB \overset{k_3}{\rightarrow} E + Y + Z$$

The concentrations of A and B are much larger than the concentrations of E. Apply the steady state treatment and obtain an expression for the rate of formation of the product.

Hint: Remember that the total enzyme concentration, $[E]_0$ is constant and equals the sum of the concentrations of enzyme in the free and complexed forms.

3) The iodine catalyzed decomposition of acetaldehyde is shown below:

$$I_2 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2I\cdot$$

$$I\cdot + CH_3CHO \overset{k_2}{\rightarrow} HI + \cdot CH_3CO$$

$$\cdot CH_3CO \overset{k_3}{\rightarrow} \cdot CH_3 + CO$$

$$\cdot CH_3 + HI \overset{k_4}{\rightarrow} CH_4 + I\cdot$$

Apply the steady state treatment and obtain an expression for the rate of formation of CO.